United States Patent	12405624
Kind Code	B2
Date of Patent	September 02, 2025
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Linear voltage regulator and test system

Abstract

A linear voltage regulator is disclosed. The voltage regulator is configured and controlled such that its output current is substantially equal to the input current. A test system for measuring electrical current consumption of a device under test (DUT) is disclosed. The test system comprises a controller coupled to a power capacitor, a voltage regulator, and a switching circuit. The controller may be configured to: control the switching circuit to place the test system into a charging state to charge the power capacitor with the DC voltage source, at a first time point, control the switching circuit to place the test system into a measurement state such that the power capacitor provides the capacitor voltage to the voltage regulator, and at a second time point after the first time point, calculate electrical current provided by the power capacitor.

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Appl. No.: **18/600016**

Filed: **March 08, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20240210980 A1	Jun. 27, 2024

Related U.S. Application Data

continuation parent-doc US 16940669 20200728 US 11960311 child-doc US 18600016

Publication Classification

Int. Cl.: **G05F1/575** (20060101); **G01R1/28** (20060101)

U.S. Cl.:

CPC **G05F1/575** (20130101); **G01R1/28** (20130101);

Field of Classification Search

CPC: G05F (1/462); G05F (1/465); G05F (1/468); G05F (1/56); G05F (1/575); G05F (1/562); G05F (1/565); G05F (1/567); G05F (1/569); G05F (1/571); G05F (1/573); G05F (1/5735); G01R (19/165); G01R (19/16504); G01R (19/16509); G01R (19/16514); G01R (19/16519); G01R (19/16523); G01R (19/16528); G01R (19/16533); G01R (19/16538); G01R (19/16542); G01R (19/16547); G01R (19/16552); G01R (19/16557); G01R (19/16561); G01R (19/16566); G01R (19/16571); G01R (19/16576); G01R (19/1658); G01R (19/16585); G01R (19/1659); G01R (19/16595); G01R (19/17); H02M (3/07); H02M (3/073); H02M (7/10); H02M (7/103); H02M (7/106); H02M (1/088); H02M (2003/071); H02M (2003/072); H02M (2003/075); H02M (2003/076); H02M (2003/077); H02M (2003/078); H02M (2001/007); H02M (2001/0048); H02M (3/1582); H02M (3/1584); H02M (3/285); H02M (3/33561); H02M (7/49); H02M (1/045); H02M (7/006); H02M (7/06); H02M (7/068); H02M (7/153); H02M (7/19); H02M (7/08); H02M (7/17); H02M (7/493); H02M (7/53806); H02M (7/5381); H02M (7/483); H02M (7/217); H02M (7/538466); H02M (7/5387); H02M (7/53871); H02M (7/53873); H02M (7/53875); H02M (1/084); H02M (1/0845); H02J (3/46); H02J (3/38)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/940,669, filed Jul. 28, 2020, titled “LINEAR VOLTAGE

REGULATOR WITH ISOLATED SUPPLY CURRENT," the entire contents of which have been herein incorporated by reference for all purposes.

TECHNICAL FIELD

(1) Embodiments of the subject matter described herein relate generally to a test system for measuring electrical current consumed by a device under test, and to a current-isolated voltage regulator suitable for use in the test system.

BACKGROUND

(2) Electronic devices, systems, and components are routinely subjected to electrical tests during manufacturing and/or in an ongoing manner after deployment. For example, an electronic device can be tested to measure the amount of electrical current it consumes during different operating modes. An unusually low or high measured current can be an indicator of a fault, error, or manufacturing defect.

(3) A low power consumption electronic device, such as a battery powered medical device, may utilize a switch-mode power supply (SMPS) that provides an as-needed switching scheme at low load currents that occur during the device's standby mode. In such a device, the switching during standby mode occurs on an as-needed basis. Accordingly, the switching period is often increased to a point where input filter capacitors become ineffective at smoothing the current. This results in a discontinuous input current that typically resembles a pulse train having high dynamic range. In this regard, the non-switching currents between "wake up" current pulses may be four to five orders of magnitude less than the switching pulses. As the switching period increases for a given pulse width (i.e., the duty cycle decreases), the overall current measurement accuracy becomes increasingly affected by the non-switching current. The combination of these factors adversely impact the effectiveness and accuracy of most readily available electrical current measurement systems, which are primarily designed to measure continuous current and/or current having a low dynamic range. As a result, measurement accuracy of discontinuous electrical current with high dynamic range suffers when such devices are tested with conventional (and economically feasible) current measurement equipment.

(4) A test system that measures electrical current may include a voltage regulator to provide operating power to the device under test. A conventional low-dropout or linear voltage regulator utilizes the regulator input voltage to source certain components, such as an internal voltage reference and an error amplifier. Consequently, the input current of this type of voltage regulator will always be higher than the output current. Although this type of voltage regulator is appropriate in some applications, it may not be suitable in certain applications where it is desirable to have the output current match the input current.

BRIEF SUMMARY

(5) Disclosed herein are a linear voltage regulator and a test system for measuring electrical current consumption of a device under test (DUT).

(6) In some embodiments, a linear voltage regulator comprises: an input voltage terminal configured to receive an input current; an output voltage terminal configured to output an output current substantially equal to the input current; a series pass field-effect transistor coupled between the input voltage terminal and the output voltage terminal; a voltage reference source to provide a reference voltage for the linear voltage regulator, the voltage reference source powered by an independent voltage supply that is isolated from the input voltage terminal; a buffer amplifier comprising a positive buffer input coupled to the output voltage terminal, and a negative buffer input coupled to a buffer output, the buffer amplifier powered by the independent voltage supply; and an error amplifier comprising an error output coupled to the series pass field-effect transistor, a positive error input operatively coupled to the output voltage terminal, and a negative error input coupled to the voltage reference source to receive the reference voltage, the error amplifier powered by the independent voltage supply, wherein the output of the error amplifier controls

impedance of the series pass field-effect transistor to adjust a regulator output voltage at the output voltage terminal.
(7) In some embodiments, a test system for measuring electrical current consumption of a device under test (DUT) may comprise: a power capacitor configured to provide a capacitor voltage; a voltage regulator comprising a regulator input terminal and a regulator output terminal; a switching circuit to regulate electrical connections between a direct current (DC) voltage source, the power capacitor, and the voltage regulator; and a controller coupled to the power capacitor, the voltage regulator, and the switching circuit. The controller may be configurable to: control the switching circuit to place the test system into a charging state to charge the power capacitor with the DC voltage source; at a first time point, control the switching circuit to place the test system into a measurement state such that the power capacitor provides the capacitor voltage to the voltage regulator; and at a second time point after the first time point, the second time point characterized by a decrease in a voltage of the power capacitor due to loading by the DUT, calculate electrical current provided by the power capacitor in a time period recorded during operation of the test system in the measurement state.
(8) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) A more complete understanding of the subject matter may be derived by referring to the detailed description and claims when considered in conjunction with the following figures, wherein like reference numbers refer to similar elements throughout the figures.
(2) FIG. **1** is a block diagram that depicts an embodiment of a test system in a typical testing environment;
(3) FIG. **2** is a schematic diagram of an embodiment of a test system connected to a device under test;
(4) FIG. **3** is a flow chart that illustrates an embodiment of an automated method of measuring electrical current of a device under test;
(5) FIG. **4** is a graph that includes plots of voltage levels over time, as sampled during a typical current measurement test performed by the test system shown in FIG. **2**; and
(6) FIGS. **5-7** are schematic diagrams of embodiments of a linear voltage regulator having isolated supply current.
DETAILED DESCRIPTION
(7) The following detailed description is merely illustrative in nature and is not intended to limit the embodiments of the subject matter or the application and uses of such embodiments. As used herein, the word “exemplary” means “serving as an example, instance, or illustration.” Any implementation described herein as exemplary is not necessarily to be construed as preferred or advantageous over other implementations. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description.
(8) Techniques and technologies may be described herein in terms of functional and/or logical block components, and with reference to symbolic representations of operations, processing tasks, and functions that may be performed by various computing components or devices. Such operations, tasks, and functions are sometimes referred to as being computer-executed, processor-based, software-implemented, computer-implemented, or the like. It should be appreciated that the

various block components shown in the figures may be realized by any number of hardware, software, and/or firmware components configured to perform the specified functions. For example, an embodiment of a system or a component may employ various integrated circuit components, e.g., memory elements, digital signal processing elements, logic elements, look-up tables, or the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices.

(9) When implemented in software or firmware, various elements of the systems described herein are essentially the code segments or instructions that perform the various tasks. In certain embodiments, the program or code segments are stored in a tangible processor-readable medium, which may include any medium that can store or transfer information. Examples of a non-transitory and processor-readable medium include an electronic circuit, a semiconductor memory device, a ROM, a flash memory, an erasable ROM (EROM), a floppy diskette, a CD-ROM, an optical disk, a hard disk, or the like.

(10) "Node"—As used herein, a "node" means any internal or external reference point, connection point, junction, signal line, conductive element, or the like, at which a given signal, logic level, voltage, data pattern, current, or quantity is present. Furthermore, two or more nodes may be realized by one physical element (and two or more signals can be multiplexed, modulated, or otherwise distinguished even though received or output at a common node).

(11) "Coupled"—The following description may refer to elements or nodes or features being "coupled" together. As used herein, unless expressly stated otherwise, "coupled" means that one element/node/feature is directly or indirectly joined to (or directly or indirectly communicates with) another element/node/feature, and not necessarily mechanically. Thus, although the schematic shown in FIG. **2** depicts one exemplary arrangement of elements, additional intervening elements, devices, features, or components may be present in an embodiment of the depicted test system. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in an embodiment of the subject matter.

(12) FIG. **1** is a block diagram that depicts an embodiment of a test system **100** in a typical testing environment that includes a device under test (DUT) **102** coupled to the test system **100** in a way that allows the test system **100** to perform one or more electrical tests on the DUT **102**. The test system **100** may be implemented as a "bench test" component having a chassis or housing that contains various devices, elements, and electronic components. In certain embodiments, the test system **100** includes a power cord **104** with a standard alternating current (AC) power plug **106** to connect with a mains power source. Alternatively or additionally, the test system **100** can receive direct current (DC) operating voltage from an external power supply **108**, which obtains AC voltage from a mains power source via a power cord **110** and AC power plug **112**. In such an arrangement, the test system **100** may include at least one input interface **114** to obtain the DC voltage from the external power supply **108** (e.g., cable plug sockets, clip terminals, a connector, or the like).

(13) The test system **100** includes at least one DUT power interface **118** that establishes an electrical connection **120** between a voltage regulator of the test system **100** and the DUT **102**. The DUT power interface **118** can be implemented in various form factors, depending on the configuration of the DUT **102**, the native power supply of the DUT **102**, the manner in which the DUT **102** is to be tested, and the like. For example, the DUT power interface **118** may include or cooperate with any of the following, without limitation: an electrical connector; cable plug sockets, clip terminals, an adapter, or the like. In certain embodiments, the DUT power interface **118** includes or cooperates with a DC voltage cord or cable that terminates with structure that emulates the shape and size of a battery that is normally used as the power source of the DUT **102** (such as a 1.5 volt AA battery). The terminating structure includes electrical contacts that simulate the

positive and negative terminals of the DUT's battery, such that the voltage regulator of the test system **100** can provide operating voltage to the DUT **102** during testing.

(14) The DUT **102** can be any electronic device having operating voltage and current specifications that are supported by the test system **100**. In this regard, the test system **100** must be able to generate sufficient DC operating voltage and current to power the DUT **102** during testing. In certain applications, the DUT **102** is a portable battery powered medical device, such as a personal insulin infusion device. In certain embodiments, the DUT **102** operates in a low power standby mode that is characterized by a discontinuous and high dynamic range standby current waveform. In accordance with a nonlimiting example, the standby current waveform includes discontinuous pulses that peak at approximately 100 mA and persist for only tens of microseconds, with intervening current of approximately 1.00 μA between the pulses, which may occur every 10-100 milliseconds. As mentioned previously, conventional current meters struggle to accurately measure current having such a high dynamic range.

(15) The test system **100** disclosed here represents an effective, low cost, and elegant solution to the problem outlined above. The test system **100** employs a method of current measurement that is based on the charging and discharging characteristics of a power capacitor having a known, calibrated capacitance. The power capacitor serves as the source of power for the DUT **102**, and the charge on the capacitor is directly linked to the current being sourced or sinked, which is defined by physics and the electrical characteristics of the capacitor. The test system **100** uses the power capacitor to measure the total current consumed by the DUT **102** during a measurement time period, independent of the DUT's current waveform type, dynamic range, etc. As explained in more detail below, the test system **100** employs a circuit configuration that is automatically controlled to isolate the current path from the power capacitor to the DUT **102**. Any leakage current or quiescent current consumed by components of the test system **100** is negligible relative to the amount of current to be measured, or is isolated such that it has no impact on the current measurement.

(16) FIG. **2** is a schematic diagram that depicts an embodiment of the test system **100** coupled to the DUT **102**. For the embodiment of FIG. **2**, all of the illustrated items (other than the DUT **102**) are part of the test system **100**. For the sake of clarity and simplicity, the power cord **104**, AC power plug **106**, input interface **114**, and DUT power interface **118** (see FIG. **1**) are not shown in FIG. **2**. The illustrated embodiment of the test system **100** includes, without limitation: a power capacitor **202**; a voltage regulator **204**; a switching circuit having a first switching element **206** and a second switching element **208**; a controller **210**; a controller clock **212**; a display device **214**; a DC voltage source **216**; one or more isolated power sources **217**; a first current-isolating buffer **218**; a second current-isolating buffer **220**; a third current-isolating buffer **222**; a diode **223**; a capacitance calibration circuit **224**; and various electrically conductive paths, traces, interconnects, or elements that serve to couple the components of the test system **100** together as needed.

(17) The first switching element **206** is coupled between the DC voltage source **216** and a regulator input terminal **230** of the voltage regulator **204**. The second switching element **208** is coupled between the DC voltage source **216** and a power terminal **232** of the power capacitor **202** (the power terminal **232** is electrically coupled to the positive conductor or plate of the power capacitor **202**, as depicted in FIG. **2**). The power capacitor **202** has a ground terminal **234** that is electrically coupled to a ground potential of the test system **100**. The diode **223** has an anode **236** coupled to the power terminal **232** of the power capacitor **202**, and a cathode **238** coupled to the regulator input terminal **230** of the voltage regulator **204**. The voltage regulator **204** has a regulator output terminal **240** that can be coupled to the DUT **102** for testing purposes. In this regard, the DUT **102** can be removably connected to the test system **100** to establish the electrical coupling between the regulator output terminal **240** and the electronics of the DUT **102**.

(18) The controller **210** is coupled to at least the power capacitor **202**, the voltage regulator **204**, the first switching element **206**, the second switching element **208**, the display device **214**, and the controller clock **212**. In accordance with the depicted implementation: the controller **210** is coupled

to the voltage regulator **204** via an analog output port or terminal **244**; the third current-isolating buffer **222** is coupled between the regulator output terminal **240** and the controller **210** via an analog input port or terminal **246**; the controller **210** is coupled to the display device via a display output interface **248**; the controller **210** is coupled to the controller clock via a clock interface **250**; the second current-isolating buffer **220** is coupled between the power terminal **232** of the power capacitor **202** and a second voltage input terminal **252** of the controller **210**; the controller **210** is coupled to the first switching element **206** via a first switch control port or terminal **254**; the controller **210** is coupled to the second switching element **208** via a second switch control port or terminal **256**; and the first current-isolating buffer **218** is coupled between the regulator input terminal **230** and a first voltage input terminal **258** of the controller **210**. In FIG. **2**, the SW**1** and SW**2** labels represent switch control signals that are used to control the switching states of the first switching element **206** and the second switching element **208**, respectively.
(19) The isolated power source(s) **217** are coupled to certain components, devices, or features of the test system **100** as appropriate to the particular embodiment. For the sake of clarity and simplicity, the various couplings associated with the isolated power source(s) are not depicted in FIG. **2**. The capacitance calibration circuit **224** may be realized as a separate circuit module (as depicted in FIG. **2**), or it may be implemented with at least some of the other components and features of the test system **100**, such as the controller **210**, the voltage regulator **204**, the diode **223**, and corresponding interconnections. For the sake of clarity and simplicity, the various couplings associated with the capacitance calibration circuit are not depicted in FIG. **2**.
(20) The DC voltage source **216** provides operating voltage(s) for the test system **100**. In certain embodiments, the test system **100** includes the DC voltage source **216**, as depicted in FIG. **2**. If internal to the test system **100**, the DC voltage source **216** can be powered by the mains power source. In some embodiments, however, the DC voltage source **216** may be external to the test system **100**. The DC voltage source **216** charges the power capacitor **202** when the second switching element **208** is closed, and provides a DC input voltage to the voltage regulator **204** when the first switching element **206** is closed. The voltage regulator **204** is configured and controlled to generate an appropriate DUT operating voltage at the regulator output terminal **240**, based on the DC input voltage that is present at the regulator input terminal **230**. Accordingly, the DC voltage source **216** provides a DC voltage that is high enough to charge the power capacitor **202** to its charged voltage level, and high enough to allow the voltage regulator **204** to generate the necessary operating voltage for the DUT **102**. In certain nonlimiting embodiments, the nominal operating voltage of the DUT **102** is 1.5 VDC, and the DC voltage source **216** provides 12.0 VDC.
(21) The switching circuit includes at least the first switching element **206** and the second switching element **208**. The switching elements **206**, **208** may be implemented as solid state (transistor-based) switches, or as electromechanical relays. Ideally, the switching elements **206**, **208** consume little to no current. Transistor-based switches are appropriate if the amount of switch leakage current is low enough to be considered negligible, relative to the expected amount of DUT current to be measured. For example, if the measured DUT current is expected to be in the range of about one nanoamp or greater, then switches having leakage current in the picoamp range may be suitable for use in the test system (such that the ratio of measurement current to leakage current is at least 1000:1). Relays typically exhibit little to no leakage current and, therefore, are suitable for use in the test system **100**.
(22) For this particular application, the capacitance of the power capacitor **202** should be stable across different working voltages, operating temperatures, environmental conditions, and the like. Accordingly, the type (composition) of the power capacitor **202** should provide a tightly controlled capacitance. For example, the power capacitor **202** may be a polypropylene film type, high voltage capacitor (e.g., generally greater than 100 volts). The capacitance can be selected to suit the needs and requirements of the particular application. For the example mentioned here (where the DC voltage source **216** provides 12 VDC, and the DUT **102** is powered by a 1.5 VDC source), the

capacitance of the power capacitor **202** can be a value within the range of about 1.0 mF to about 10.0 mF (for smaller, low power devices). These capacitor sizes are readily available in polypropylene film.

(23) The diode **223** is a passive component that may be a conventional off-the-shelf item. The diode **223** has very low reverse leakage, which should be in the range of about 1,000 times less than the intended measurement current. Accordingly, a silicon diode is best suited for this application (rather than a Schottky diode). The diode **223** does not consume any measurable quiescent current and, therefore, it can appear in the measured current flow path (as depicted in FIG. **2**).

(24) The voltage regulator **204** is digitally controlled by the controller **210** such that the current consumed by the voltage regulator **204** is isolated, separated, or otherwise not considered in the measurement of the DUT current. The controller samples the regulator output voltage (at the analog input terminal **246**) and generates an appropriate control signal (at the analog output terminal **244**) to increase or decrease the regulator output voltage as needed in an ongoing manner. As explained in more detail below, the voltage regulator **204** is sourced by the isolated power source(s) **217** rather than by the DC voltage that appears at the regulator input terminal **230**. Consequently, the quiescent current consumed by the voltage regulator **204** is associated with the isolated power source(s) **217**, and the voltage regulator **204** operates in a current-isolated manner relative to the current consumed by the DUT **102** during testing. Additional details of the voltage regulator **204** are described below with reference to FIGS. **5-7**.

(25) Each of the current-isolating buffers **218**, **220**, **222** may be implemented as a unity gain operational amplifier having a conventional layout and configuration. Although not shown in FIG. **2**, the current-isolating buffers **218**, **220**, **222** may include or cooperate with a simple voltage divider circuit if needed for compatibility with the analog inputs of the controller **210**. The current-isolating buffers **218**, **220**, **222** allow the controller **210** to sample the respective voltages, without consuming any measurable current. The current-isolating buffers **218**, **220**, **222** are powered by the isolated power source(s) **217**, and they have very low input leakage current. The input leakage current is low enough to make it negligible relative to the amount of DUT current that is to be measured. Thus, the measured DUT current remains accurate and precise even though the current-isolating buffers **218**, **220**, **222** branch off of the measurement current flow path. Stated another way, the buffers **218**, **220**, **222** are suitably configured and arranged to isolate the controller **210** from the test current flow path between the power capacitor **202** and the DUT **102**.

(26) The test system **100** calculates the electrical current consumed by the DUT **102** during a measurement period of time, and generates the calculated current as an output. The display device **214** represents one type of output device that can be used to display the calculated current as an output. The display device **214** may be integrated with the housing or chassis of the test system **100**, or it may be realized as a separate peripheral component that connects to and/or communicates with the test system **100**. Any type of display technology and form factor can be utilized with the display device **214**, and the specific implementation details of the display device **214** will not be described here. In addition to, or instead of, the display device **214**, the test system **100** may include or cooperate with other output devices or systems, such as a printer, an audio transducer, a mechanical output device, or an interface that sends a notification, an email, a text message, an electronic report, or the like.

(27) The controller **210** may be realized as one or more physical devices, such as a microcontroller unit, a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), a system on a chip (SOC), or the like. In certain embodiments, the controller **210** is realized as a "single device" microcontroller unit that includes a processor core (e.g., a CPU), a storage medium for processor executable program instructions, an input/output interface, at least one digital-to-analog converter (DAC), at least one analog-to-digital converter (ADC), memory (volatile and nonvolatile), and other peripheral components or elements as needed. The controller **210** may be

based on an off-the-shelf component that is programmed, configured, and/or customized for use in the test system **100**. To this end, the controller **210** is configurable to carry out the various processes, methods, operations, and functions described herein.

(28) As mentioned previously, the test system **100** is designed, configured, and operated such that the current consumed by the DUT **102** can be accurately and precisely measured in an isolated manner. To this end, leakage current, quiescent current, and/or operating current consumed by certain components, devices, and elements of the test system **100** are minimized to the point where their contribution is negligible, or the sources of such current are isolated from the test current flow path. In this regard, the isolated power source(s) **217** may be utilized to provide operating voltage to one or more of the following items: the first switching element **206**; the second switching element **208**; the voltage regulator **204**; the controller **210**; the display device **214**; the controller clock **212**; the first current-isolating buffer **218**; the second current-isolating buffer **220**; the third current-isolating buffer **222**; and the capacitance calibration circuit **224**.

(29) As explained in more detail below, the known capacitance of the power capacitor **202** is used to calculate the current consumed by the DUT **102** during the measurement time period. Although the power capacitor **202** is chosen such that its capacitance is relatively stable and constant, it can still be susceptible to slight variation over time. Thus, it is important to have an accurate calibrated capacitance value. The capacitance calibration circuit **224** is couplable to the power capacitor **202** to calibrate the capacitance of the power capacitor **202**. Calibration may occur whenever the test system **100** is powered up, before each current measurement, daily, weekly, or the like. The capacitance calibration circuit **224** can be implemented to self-calibrate the test system **100** using, for example, a constant and known current source, which may cooperate with other components of the test system **100** to perform a calibration routine. When performing a calibration, the fixed current source (e.g., a constant 1.0 mA current) takes the place of the DUT **102**. For calibration, the unknown variable is the capacitance, which can be calculated based on the discharge characteristics of the power capacitor **202**. The calibrated capacitance can be saved for use as a known value for subsequent current measurements, where the unknown variable is the current consumed by the DUT **102**. The test system **100** itself may also be calibrated as needed, such as annually, monthly, or the like. Calibration of the test system **100** may require external calibration equipment to calibrate the voltage sources, the fixed current source that is used to obtain the calibrated capacitance, the ADCs and DACs of the controller **210**, etc.

(30) The switching circuit is configured and controlled by the controller **210** to regulate electrical connections between the DC voltage source, the power capacitor **202**, and the voltage regulator **204**. In this regard, the controller **210** is configurable to control the switching circuit by independently opening and closing the switching elements **206**, **208** as needed. The controller **210** activates or actuates the switching elements **206**, **208** to place the test system **100** into different states or operating modes including, without limitation: a charging state; a measurement state; and a post-measurement state.

(31) For the charging state, the controller **210** keeps the first and second switching elements **206**, **208** closed to charge the power capacitor **202** with source voltage provided by the DC voltage source **216**. When both of the switching elements **206**, **208** are closed, the diode **223** is not forward biased and, therefore, current does not flow through the diode **223**. Thus, the voltage of the power capacitor **202** can be sampled and monitored by the controller **210** via the second voltage input terminal **252**. While operating in the charging state, the source voltage of the DC voltage source **216** is present at the regulator input terminal **230**, which enables the voltage regulator **204** to generate a regulated DUT operating voltage for the DUT **102**. The DUT operating voltage can operate the DUT **102** while the power capacitor **202** is being charged. Accordingly, the DUT **102** can be initialized, prepared for testing, placed into its low current standby mode, or the like, while being sourced by the DC voltage source **216**.

(32) For the measurement state, the controller **210** keeps the first and second switching elements

206, **208** open to provide the capacitor voltage to the regulator input terminal **230**. Opening the switching elements **206**, **208** isolates the DC voltage source **216** from the other components of the test system **100**. Thus, while in the measurement state, the power capacitor **202** functions as the voltage source instead of the DC voltage source **216**—the power capacitor **202** provides its capacitor voltage to the voltage regulator **204** via the diode **223**. When both of the switching elements **206**, **208** are open, the diode **223** is forward biased by the capacitor voltage and, therefore, the diode permits discharge of the power capacitor **202** by way of a test current flow path **270**. The test current flow path (depicted in dashed lines) runs from the power capacitor **202**, through the diode **223**, through the voltage regulator **204**, and to the DUT **102**, which represents the electrical load that consumes the power provided by the power capacitor **202**. While operating in the measurement state, the capacitor voltage is present at the regulator input terminal **230**, which enables the voltage regulator **204** to generate the regulated DUT operating voltage for the DUT **102** (assuming that the capacitor voltage remains high enough). While operating in the measurement state, the capacitor voltage can be sampled by the controller **210** (via the second voltage input terminal **252**), and the voltage present at the regulator input terminal **230** can be sampled by the controller **210** (via the first voltage input terminal **258**).

(33) For the post-measurement state, the controller **210** keeps the first switching element **206** closed and the second switching element **208** open. While the test system **100** is in the post-measurement state, the DC voltage source **216** provides its voltage to the regulator input terminal **230** and to the cathode **238** of the diode **223**, via the first switching element **206**. In the post-measurement state, the diode **223** is reverse biased and, therefore, inhibits further discharge of the power capacitor **202**. Accordingly, the source voltage generated by the DC voltage source **216** can be sampled by the controller **210** (via the first voltage input terminal **258**), and the discharged voltage of the power capacitor **202** can be sampled by the controller **210** (via the second voltage input terminal **252**) when the test system **100** is in the post-measurement state.

(34) Operation of the test system **100** will now be described with reference to FIG. **3**, which is a flow chart that illustrates an embodiment of an automated current measurement process **300**. The process **300** is performed by the test system **100** to measure electrical current consumed by the DUT **102**. The description of the process **300** may refer to elements mentioned above in connection with FIGS. **1** and **2**. It should be appreciated that the process **300** may include any number of additional or alternative tasks, the tasks shown in FIG. **3** need not be performed in the illustrated order, and the process **300** may be incorporated into a more comprehensive procedure or process having additional functionality not described in detail herein. Moreover, one or more of the tasks shown in FIG. **3** could be omitted from an embodiment of the process **300** as long as the intended overall functionality remains intact.

(35) The process **300** may begin by calibrating the capacitance of the power capacitor **202** (task **302**). As explained above, calibration need not be performed for each measurement and, therefore, task **302** may be performed periodically, in accordance with a particular schedule, or the like. Nonetheless, task **302** is shown for the sake of completeness. The following description of the process **300** assumes that the test system **100** has an accurate calibrated value of the capacitance, which can be used to calculate the amount of current consumed by the DUT **102** during the measurement period. The DUT **102** is connected to the test system **100** in an appropriate manner (task **304**) to establish an electrical coupling between the regulator output terminal **240** and the electronics of the DUT **102**. In this way, the voltage regulator **204** can serve as the power source of the DUT **102**.

(36) After the DUT **102** has been connected to the test system **100**, the current measurement routine begins. The test may begin automatically in response to connecting the DUT **102**, or it may require a user instruction or command. In certain embodiments, the test system **100** automatically controls the switching circuit with the controller **210** to place the test system into the charging state (task **306**). As mentioned above, the controller **210** keeps the first and second switching elements **206**,

208 closed while the test system **100** is in the charging state, such that the DC voltage source **216** charges the power capacitor **202**. Moreover, the DC voltage source **216** provides an input voltage to the voltage regulator **204**, which in turn provides an appropriate operating voltage to the DUT **102**. Accordingly, the DUT **102** can be initialized and otherwise prepared for the current measurement routine.

(37) The charging state is maintained until the power capacitor **202** is charged (e.g., the capacitor voltage has reached a charged voltage level). If the power capacitor **202** has not reached the charged voltage (the “No” branch of query task **308**), then the test system remains in the charging state. If the power capacitor **202** has reached the charged voltage (the “Yes” branch of query task **308**), then the process **300** continues by automatically controlling the switching circuit with the controller **210** to transition the test system **100** from the charging state into the measurement state (task **310**). In some embodiments, query task **308** involves comparing the capacitor voltage against a charged voltage threshold, such that the switching circuit is automatically controlled to place the test system **100** into the measurement state when the capacitor voltage reaches the charged voltage threshold. The controller **210** can sample the capacitor voltage at the second voltage input terminal **252** for purposes of this comparison. In some embodiments, query task **308** involves monitoring elapsed time after entering the charging state, such that the switching circuit is automatically controlled to place the test system into the measurement state when the elapsed time exceeds a charging time threshold. The controller **210** can maintain a counter or a timer (based on operation of the controller clock **212**) to monitor the elapsed time.

(38) This example assumes that the power capacitor **202** has reached its charged voltage and that the test system **100** has transitioned to the measurement state. As mentioned above, the controller **210** opens the first and second switching elements **206**, **208** to place the test system **100** into the measurement state, and keeps them open while the test system operates in the measurement state. For the measurement state, the DC voltage source **216** is disconnected from the remaining components, and the power capacitor **202** provides its capacitor voltage to the voltage regulator **204**. In response to the transition to the measurement state, the controller **210** begins (or resumes) sampling the voltages at the first voltage input terminal **258** and the second voltage input terminal **252** (task **312**), and records a measurement start time (task **314**). In certain embodiments, tasks **310**, **312**, and **314** are performed concurrently such that the measurement start time is recorded, the voltages are sampled, and the switching elements **206**, **208** are opened at the same time. Thus, the measurement start time is associated with opening of the switching elements **206**, **208**.

(39) The measurement state is maintained until the power capacitor **202** has reached a discharged voltage in response to loading, i.e., operation of the DUT **102**. In this regard, the capacitor voltage drops over time due to current consumed by the DUT **102**. If the power capacitor **202** has not reached the discharged voltage (the “No” branch of query task **316**), then the test system **100** remains in the measurement state. If the power capacitor **202** has reached the discharged voltage (the “Yes” branch of query task **316**), then the process **300** continues by automatically controlling the switching circuit with the controller **210** to transition the test system **100** from the measurement state into the post-measurement state (task **318**). The test system **100** is designed and operated such that the power capacitor **202** is not discharged too much during the measurement state, to ensure that the voltage characteristics of the power capacitor remain linear.

(40) In some embodiments, query task **316** involves comparing the capacitor voltage against a minimum capacitor voltage threshold, such that the switching circuit is automatically controlled to place the test system **100** into the post-measurement state when the capacitor voltage is less than or equal to the minimum capacitor voltage threshold. The controller **210** can sample the capacitor voltage at the second voltage input terminal **252** for purposes of this comparison. For the example presented here, where the DC voltage source **216** provides 12 VDC, the minimum capacitor voltage threshold may be about 8 VDC. In some embodiments, query task **316** involves monitoring elapsed time after entering the measurement state (i.e., after opening of the switching elements **206**, **208**),

such that the switching circuit is automatically controlled to place the test system **100** into the post-measurement state when the elapsed time is greater than or equal to a maximum time threshold. The controller **210** can maintain a counter or a timer (based on operation of the controller clock **212**) to monitor this elapsed time. For the example presented here, the maximum time threshold may be on the order of one to two seconds.

(41) This example assumes that the power capacitor **202** has reached the discharged voltage and that the test system **100** has transitioned to the post-measurement state. As mentioned above, the controller **210** closes the first switching element **206** and keeps the second switching element **208** open to place the test system **100** into the post-measurement state, and maintains those switch conditions while the test system **100** operates in the post-measurement state. For the post-measurement state, the DC voltage source **216** is coupled to: the cathode **238** of the diode **223**; the input of the first current-isolating buffer **218**; and the regulator input terminal **230**. The open state of the second switching element **208**, however, keeps the DC voltage source **216** disconnected from: the power capacitor **202**; the anode **236** of the diode **223**; and the input of the second current-isolating buffer **220**. Accordingly, the diode **223** is reverse biased, the power capacitor **202** no longer discharges, and the capacitor voltage (which is sampled at the second voltage input terminal **252**) remains stable in the post-measurement state. Moreover, the voltage provided by the DC voltage source **216**, which corresponds to the charged voltage of the power capacitor **202**, can be sampled at the first voltage input terminal **258** in the post-measurement state.

(42) In response to the transition to the post-measurement state, the controller **210** records a measurement end time (task **320**). In certain embodiments, tasks **318** and **320** are performed concurrently such that the first switching element **206** is closed and the measurement end time is recorded at the same time. Thus, the measurement end time is associated with closing of the first switching element **206**. The controller **210** may stop sampling the voltages at the first voltage input terminal **258** and the second voltage input terminal **252** (task **322**) at any suitable time following the transition to the post-measurement state. For reasons explained below, the controller **210** continues sampling these voltages in the post-measurement state for a period of time, to ensure that the voltages have stabilized.

(43) The process **300** continues by calculating the electrical current provided by the power capacitor **202** (and consumed by the DUT **102**) in a time period recorded during operation of the test system **100** in the measurement state (task **324**). For this example, the time period is defined by the recorded measurement start time and the recorded measurement end time, and the calculation is based on a sampled value of the charged voltage of the power capacitor **202**, a sampled value of the discharged voltage of the power capacitor **202**, and discharge characteristics of the power capacitor **202**—namely, the relationship between capacitance and capacitor voltage over time with respect to electrical current. More specifically, the controller **210** calculates the electrical current consumed by the DUT **102** between the recorded measurement start time and the recorded measurement end time in accordance with the expression

(44) $i = C\frac{V_c - V_d}{t_f - t_i}$,

where: C is the known (calibrated) capacitance of the power capacitor; V.sub.c is the sampled value of the charged voltage; V.sub.d is the sampled value of the discharged voltage; t.sub.i is the recorded measurement start time; and t.sub.f is the recorded measurement end time. The controller **210** records the measurement start and end times, samples the capacitor voltage at the second voltage input terminal **252**, and samples the input voltage of the voltage regulator **204** at the first voltage input terminal **258**. Thus, the current consumed by the DUT **102** can be easily determined at task **324**.

(45) The process **300** continues by generating the calculated electrical current as an output of the test system **100** (task **326**). For example, the display device **214** can be controlled and driven in an appropriate manner to display the calculated electrical current in any desired format, e.g., a numerical readout. For this particular example, the test system **100** displays the standby current

consumed by the DUT **102** during the measurement time period, which is typically several seconds. The standby current represents an average of the instantaneous current measured over the recorded time period.

(46) FIG. **4** is a graph that includes plots of voltage levels over time, as sampled during a typical current measurement test performed by the test system **100** shown in FIG. **2**. The horizontal time axis indicates the measurement start time (t.sub.i) and the measurement end time (t.sub.f). The vertical voltage axis indicates the charged voltage (V.sub.c) and the discharged voltage (V.sub.d) of the power capacitor **202**. In FIG. **4**, the dashed line plot **402** corresponds to the voltage present at the regulator input terminal **230**, the cathode **238** of the diode **223**, and the input of the first current-isolating buffer **218**. In other words, the plot **402** represents the voltage sampled at the first voltage input terminal **258** of the controller **210**. The solid line plot **404** corresponds to the voltage present at the power terminal **232** of the power capacitor **202**, the anode **236** of the diode **223**, and the input of the second current-isolating buffer **220**. In other words, the plot **404** represents the voltage sampled at the second voltage input terminal **252** of the controller **210**. The two plots **402**, **404**, track each other during the charging state (before the measurement start time) and during the measurement state (between the measurement start time and the measurement end time), due to the status of the switching elements **206**, **208** during that period of time. The two plots **402**, **404** diverge at the measurement end time, due to the closure of the first switching element **206** (the second switching element **208** remains open). As explained above, the transition from the measurement state to the post-measurement state reconnects the DC voltage source **216** to the voltage regulator **204**, which makes the source voltage (i.e., the charged voltage) immediately available for sampling at the first voltage input terminal **258**. The capacitor voltage, however, remains stable in the post-measurement state. Accordingly, the discharged voltage of the power capacitor **202** is available for sampling at the second voltage input terminal **252**.

(47) For the embodiment shown in FIG. **2**, it is assumed that the charged voltage of the power capacitor **202** equals the voltage generated by the DC voltage source **216**. Accordingly, the charged voltage value may be sampled at the first voltage input terminal **258** at a sampling time that occurs before the measurement start time and/or at a sampling time that occurs after the measurement end time (i.e., while the test system **100** is in the post-measurement state). Alternatively or additionally, the charged voltage value may be sampled at the second voltage input terminal **252** at a sampling time that occurs before the measurement start time.

(48) The embodiment of the test system **100** shown in FIG. **2** samples the discharged voltage at the second voltage input terminal **252** at a sampling time that occurs after the measurement end time, while the capacitor voltage remains constant. For example, the test system **100** may wait a number of sampling periods or a designated amount of time before sampling the discharged voltage, or it may continue to sample the discharged voltage level but only consider sampled values obtained after the designated amount of time (e.g., 100 ms, one second, or the like). This sampling scheme takes into account the equivalent series resistance (ESR) of the power capacitor **202**. The effect of ESR on the sampled voltages is schematically depicted in FIG. **4**. At the measurement start time, both plots **402**, **404** exhibit a sudden voltage drop. This drop occurs when the switching elements **206**, **208** are opened to insert the power capacitor **202** into the current flow path **270**; the voltage drop is caused by the ESR of the power capacitor **202**. At the measurement end time, however, the power capacitor **202** is removed from the current flow path **270**. Consequently, the ESR of the power capacitor **202** results in a quick voltage recovery before the capacitor voltage stabilizes to its discharged voltage level. The controller **210** uses a sampled value of the stabilized capacitor voltage to calculate the current consumed by the DUT **102**.

(49) Referring again to task **324** and the expression used to calculate the measured current, the charged voltage (V.sub.c) may be sampled: at the first voltage input terminal **258** of the controller **210** at a sampling time that occurs before the measurement start time; at the first voltage input terminal **258** of the controller **210** at a sampling time that occurs after the measurement end time;

and/or at the second voltage input terminal **252** of the controller **210** at a sampling time that occurs before the measurement start time. The discharged voltage (V.sub.d) is sampled at the second voltage input terminal **252** at a sampling time that occurs after the measurement end time. Although not required, the charged voltage and the discharged voltage can be sampled at the same sampling time (while the test system **100** is in the post-measurement state).
(50) The current measurement procedure described here can reliably and accurately measure the average current consumed by a DUT, even when the current exhibits a very high dynamic range and relatively low peak current values. The test system **100** described here can be fabricated from inexpensive and readily available parts and components, and can be modified as needed to effectively support different types of DUTs having different functional specifications, voltage requirements, and current consumption characteristics.
(51) As mentioned above, the voltage regulator **204** is suitably configured and controlled such that its output current very closely matches its input current (i.e., any quiescent current consumed by the voltage regulator **204** is negligible relative to the amount of DUT current being measured). Linear voltage regulators are used to produce a constant output voltage from a varying input voltage of higher magnitude. Linear voltage regulators typically include a series pass transistor (which may be a MOSFET or a BJT), an error amplifier, a voltage reference, and a feedback divider network. The voltage reference establishes a fixed voltage value that is used for comparison purposes. The feedback divider network produces a scaled version of the output voltage with a reduction ratio equal to the desired output voltage divided by the reference voltage. The error amplifier compares the scaled output voltage against the reference voltage, and drives the series pass transistor to vary its resistance. Ideally, the transistor is driven to minimize the difference between the output voltage and the reference voltage.
(52) A typical linear voltage regulator that follows conventional design methodologies includes three terminals: an input voltage terminal (VIN); an output voltage terminal (VOUT); and a ground terminal (GND). The input voltage is applied between the VIN and GND terminals, and the regulated output voltage is supplied between the VOUT and GND terminals. The series pass transistor places the VIN and VOUT terminals in a series circuit. Theoretically, the current flowing into the VIN terminal (I.sub.IN) should be equal to the current flowing out of the VOUT terminal (I.sub.OUT). However, I.sub.IN is always higher than I.sub.OUT (for such conventional voltage regulators) because the regulator inherently consumes an amount of current that is associated with operation of the voltage reference, the error amplifier, and the feedback divider network. Depending on the topology and implementation of the voltage regulator, the magnitude of the difference between I.sub.IN and I.sub.OUT is highly variable and may also vary depending on the applied load.
(53) In some applications (e.g., the test system **100** described above), it may be desirable to have I.sub.IN match I.sub.OUT. For example, such a scenario exists when it is necessary to precisely measure the applied load current without affecting the voltage on the VOUT terminal. If the I.sub.IN is identical to I.sub.OUT, then a shunt resistor can be placed in series with the VIN terminal, and the voltage drop across the shunt resistor will be directly proportional to the magnitude of the I.sub.OUT current. In this situation, the voltage drop across the shunt resistor will not affect the regulated output voltage of the voltage regulator, because the regulator will naturally account for the drop on the VIN terminal.
(54) In accordance with certain embodiments described here, the current consumed by the voltage regulator itself is isolated from the current supplied to the load. The current-isolating design and configuration of the voltage regulator results in matching values of I.sub.IN and I.sub.OUT, wherein any difference is negligible or de minimis relative to the magnitude of I.sub.IN and I.sub.OUT. In this regard, FIG. **2** depicts an implementation where the controller **210** controls the voltage regulator **204** without consuming current from the power capacitor **202**. Moreover, the voltage regulator **204**, the controller **210**, and the third current-isolating buffer **222** obtain operating

voltage from the isolated power source(s) **217** rather than from the power capacitor **202**. FIGS. **5-7** depict embodiments of a linear voltage regulator having isolated supply current. The test system **100** can be modified (if needed) to use the voltage regulators shown in FIGS. **5-7**. FIG. **5** depicts a linear voltage regulator **500** that is based on an analog design, FIG. **6** depicts a linear voltage regulator **600** that includes or cooperates with an ADC, a DAC, and a processing core, and FIG. **7** depicts a linear voltage regulator **700** that includes or cooperates with a DAC.

(55) Referring to FIG. **5**, the linear voltage regulator **500** generally includes, without limitation: a series pass transistor **502**; an error amplifier **504**; a voltage reference source **506** to provide a reference voltage for the voltage regulator **500**; a buffer amplifier **508**; a feedback divider network that includes a first resistor **510** and a second resistor **512**; an input voltage terminal **514** (labeled VIN) for the regulator input voltage; an output voltage terminal **516** (labeled VOUT) for the regulator output voltage; a supply voltage terminal **518** (labeled VSPLY); and a ground terminal **520** (labeled GND).

(56) To achieve appropriate current isolation, the series pass transistor **502** is a metal-oxide-semiconductor field-effect transistor (MOSFET). For the depicted embodiment, the transistor **502** is a p-channel enhancement mode MOSFET. The transistor **502** is coupled between the input voltage terminal **514** and the output voltage terminal **516**. More specifically, the source of the transistor **502** is coupled to the input voltage terminal **514**, the drain of the transistor **502** is coupled to the output voltage terminal **516**, and the gate of the transistor is coupled to the error output **530** of the error amplifier **504**. The feedback divider network is coupled between the buffer output **548** and the ground terminal **520**. A negative error input **532** of the error amplifier **504** is coupled to a positive terminal **534** of the voltage reference source **506**, and a positive error input **536** of the error amplifier **504** is coupled to the feedback divider network. More specifically, the positive error input **536** is coupled between the first and second resistors **510**, **512**, which are connected in series with one another. Notably, the error amplifier **504** is powered by an independent voltage supply, which can be coupled between the supply voltage terminal **518** and the ground terminal **520** (the lines leading from the error amplifier **504** to the supply voltage terminal **518** and the ground terminal **520** represent the supply voltage connection). A negative terminal **540** of voltage reference source **506** is coupled to the ground terminal **520**, and the voltage reference source **506** is powered by the independent voltage supply (the voltage reference source **506** is coupled to the supply voltage terminal **518** to obtain the supply voltage).

(57) The buffer amplifier **508** is configured as a unity gain follower having a high input impedance. The positive buffer input **544** of the buffer amplifier **508** is coupled to the drain of the transistor **502** and to the output voltage terminal **516**. The negative buffer input **546** of the buffer amplifier **508** is coupled to the buffer output **548** of the buffer amplifier **508** and to a first end **550** of the first resistor **510**. Notably, the buffer amplifier **508** is powered by the independent voltage supply (the lines leading from the buffer amplifier **508** to the supply voltage terminal **518** and the ground terminal **520** represent the supply voltage connection). The second end **552** of the first resistor **510** is coupled to a first end **554** of the second resistor **512** and to the positive error input **536** of the error amplifier **504**, as mentioned above. The second end **556** of the second resistor **512** is coupled to the ground terminal **520**, thus establishing the feedback divider network. The resistor values are chosen such that the feedback divider network provides a scaled output voltage (that ideally matches the reference voltage) at the divider output, which corresponds to the node defined by the second end **552** of the first resistor **510**, the first end **554** of the second resistor **512**, and the positive error input **536** of the error amplifier **504**.

(58) The basic operating principle of the linear voltage regulator **500** is similar to that described above for a traditional three-terminal voltage regulator. In this regard, the output of the error amplifier **504** controls the impedance of the transistor **502** to adjust the regulator output voltage that appears at the output voltage terminal **516**. The output of the error amplifier **504** is produced based on the difference between the reference voltage that appears at the negative error input **532**

and the scaled output voltage that appears at the positive error input **536**. For example, if the scaled output voltage is higher than the reference voltage, then the output voltage of the error amplifier **504** can be adjusted to incrementally increase the series pass resistance of the transistor **502** to reduce the regulator output voltage. Conversely, if the scaled output voltage is lower than the reference voltage, then the output voltage of the error amplifier **504** can be adjusted to incrementally decrease the series pass resistance of the transistor **502** to increase the regulator output voltage.

(59) The voltage regulator **500** achieves supply current isolation by way of the buffer amplifier **508**. The buffer amplifier **508** is configured and arranged to operate as a unity gain (1:1) follower, such that the voltage at the buffer output **548** matches the voltage at the positive buffer input **544** (the output of the buffer amplifier **508**, rather than the transistor **502**, drives the feedback divider network). Current isolation is further achieved through the use of the supply voltage terminal **518** and the ground terminal **520**, which are coupled to provide source voltage and operating current to the error amplifier **504**, the voltage reference source **506**, and the buffer amplifier **508**—these components are powered by an independent voltage supply that is coupled to the supply voltage terminal **518**, which is isolated from the input voltage terminal **514**. Accordingly, the input voltage terminal **514** supplies current only to the output voltage terminal **516**, via the transistor **502**. The inherently high input impedance of the buffer amplifier **508** results in only a negligible amount of current drawn from the output voltage terminal **516**.

(60) The transistor **502** is realized as a MOSFET (either NMOS or PMOS, although FIG. **2** depicts a PMOS implementation) to take advantage of the extremely low gate-source and gate-drain currents. A MOSFET operating in near steady-state conditions has negligible gate-source and gate-drain currents, relative to the amount of series pass current. Furthermore, the high input impedance of the positive buffer input **544** of the buffer amplifier **508** ensures that the current flow into the positive buffer input **544** is negligible compared to the desired current measurement resolution (e.g., less than 1:1000). In this regard, the buffer amplifier **508** isolates the current flow path from the input voltage terminal **514** to the output voltage terminal **516**, such that the ratio of current flowing into the positive buffer input **544** to current flowing in the current flow path is less than 1:1000. As an example, if the desired current measurement resolution is 1.0 μA, the input impedance of the buffer amplifier **508** should be on the order of 1.0 GΩ or higher, such that the input current is 1.0 nA or less.

(61) The independent voltage supply (which feeds the supply voltage terminal **518**) supplies the quiescent current needed to operate the error amplifier **504**, the voltage reference source **506**, and the buffer amplifier **508**. This arrangement provides additional current isolation because the quiescent current is separate and distinct from the current that flows through the transistor **502**.

(62) Referring to FIG. **6**, the voltage regulator **600** generally includes, without limitation: a series pass transistor **602**; a buffer amplifier **608**; a feedback divider network that includes a first resistor **610** and a second resistor **612**; an input voltage terminal **614** (labeled VIN); an output voltage terminal **616** (labeled VOUT); a supply voltage terminal **618** (labeled VSPLY); a ground terminal **620** (labeled GND); an ADC **670**; a digital processing core **672**; and a DAC **674**. The arrangement and configuration of the voltage regulator **600** are similar to that described above for the voltage regulator **500**. The illustrated embodiment of the voltage regulator **600**, however, replaces the error amplifier **504** with the combination of the ADC **670**, the processing core **672**, and the DAC **674**. Accordingly, for the sake of brevity and convenience, common or equivalent aspects of the voltage regulators **500**, **600** will not be redundantly described in detail with reference to FIG. **6**.

(63) The series pass transistor **602** is realized as a MOSFET having its source coupled to the input voltage terminal **614**, its drain coupled to the output voltage terminal **616**, and its gate coupled to an analog output **630** of the DAC **674**. An analog voltage input **636** of the ADC **670** is coupled to the divider output of the feedback divider network to receive a scaled output voltage produced by the feedback divider network. More specifically, the analog voltage input **636** is coupled between

the first and second resistors **610**, **612**, which are connected in series with one another. The positive buffer input **644** of the buffer amplifier **608** is coupled to the drain of the transistor **602** and to the output voltage terminal **616**. The negative buffer input **646** of the buffer amplifier **608** is coupled to the buffer output **648** of the buffer amplifier **608** and to a first end **650** of the first resistor **610**. The second end **652** of the first resistor **610** is coupled to a first end **654** of the second resistor **612** and to the analog voltage input **636** of the ADC **670**, as mentioned above. The second end **656** of the second resistor **612** is coupled to the ground terminal **620**, thus establishing the feedback divider network.

(64) Although depicted as separate blocks in FIG. **6**, the ADC **670**, the processing core **672**, and the DAC **674** can be combined into a single device or component, depending on the desired implementation. For example, the ADC **670**, the processing core **672**, and the DAC **674** can be realized as features or elements of a microcontroller device. For the illustrated embodiment, the ADC **670** is coupled to the processing core **672** in an appropriate manner to communicate digital information to the processing core **672**. Likewise, the processing core **672** is coupled to the DAC **674** in an appropriate manner to communicate digital information to the DAC **674**. In this regard, the ADC **670** has a digital output interface **680**, which is coupled to a digital input interface **682** of the processing core **672**. Moreover, the processing core **672** has a digital output interface **684**, which is coupled to a digital input interface **686** of the DAC **674**.

(65) The DAC **674** cooperates with a voltage reference source **688**, which provides a reference voltage for the digital-to-analog conversion. Similarly, the ADC **670** cooperates with a voltage reference source **690**, which provides a reference voltage for the analog-to-digital conversion. The reference voltage used by the DAC **674** will typically be higher than the reference voltage used by the ADC **670**. Accordingly, the voltage reference source **688** can be distinct and separate from the voltage reference source **690** (as shown). The ADC **670**, the processing core **672**, the DAC **674**, the buffer amplifier **608**, the voltage reference source **688**, and the voltage reference source **690** are powered by an independent voltage supply (as described above). Accordingly, these components are coupled to the supply voltage terminal **618** and the ground terminal **620**. For simplicity and clarity, FIG. **6** does not show separate connections between the supply voltage terminal **618** and the voltage reference sources **688**, **690**.

(66) The basic operating principle of the linear voltage regulator **600** is similar to that described above for the voltage regulator **500**. The feedback divider network provides the scaled output voltage to the ADC **670**. The digital output interface **680** of the ADC **670** provides a digital representation of the scaled output voltage that appears at the analog voltage input **636**. The processing core **672** receives the digital representation of the scaled output voltage by way of its digital input interface **682**. The processing core **672** implements an algorithm that attempts to minimize the difference between the sampled voltage and a specified reference voltage. In this regard, the processing core **672** calculates a difference between the digital representation of the scaled output voltage and a digital representation of a programmed, stored, or otherwise designated reference voltage. The processing core **672** generates a digital control output at its digital output interface **684**, wherein the digital control output is based on the calculated difference. The digital control output is provided to the DAC **674**, by way of the digital input interface **686**.

(67) The DAC **674** is configured and operated to convert the digital control output into a corresponding analog control voltage, which appears at the analog output **630** of the DAC **674**. The analog control voltage controls the impedance of the transistor **602** to adjust the regulator output voltage that appears at the output voltage terminal **616**.

(68) As described above with reference to the voltage regulator **500**, the transistor **602**, the buffer amplifier **608**, and the independent voltage supply (that feeds the supply voltage terminal **618**) cooperate to provide current isolation for the voltage regulator **600**. Accordingly, the input voltage terminal **614** supplies current only to the output voltage terminal **616**, via the transistor **602**.

(69) Referring to FIG. **7**, the voltage regulator **700** generally includes, without limitation: a series

pass transistor **702**; an error amplifier **704**; an input voltage terminal **714** (labeled VIN) for the regulator input voltage; an output voltage terminal **716** (labeled VOUT) for the regulator output voltage; a supply voltage terminal **718** (labeled VSPLY) for an independent voltage supply; a ground terminal **720** (labeled GND); and a reference voltage terminal **721** (labeled VREF) for a reference voltage. Certain aspects of the voltage regulator **700** are similar to that described above for the voltage regulators **500**, **600**. Accordingly, for the sake of brevity and convenience, common or equivalent aspects of the voltage regulators **500**, **600**, **700** will not be redundantly described in detail with reference to FIG. **7**.

(70) The series pass transistor **702** is realized as a MOSFET having its source coupled to the input voltage terminal **714**, its drain coupled to the output voltage terminal **716**, and its gate coupled to the error output **730** of the error amplifier **704**. A negative error input **732** of the error amplifier **704** is coupled to the reference voltage terminal **721** to receive the reference voltage. In certain embodiments, a positive error input **736** of the error amplifier **704** is directly connected to the drain of the transistor **702** and to the output voltage terminal **716** (in other words, no intervening components, devices, or elements are arranged between the output voltage terminal **716** and the positive error input **736**). Thus, the error amplifier **704** directly receives the regulator output voltage in a feedback path. The error amplifier **704** is powered by an independent voltage supply, which can be coupled between the supply voltage terminal **718** and the ground terminal **720**. As mentioned previously, the independent voltage supply is isolated from the input voltage terminal **714**.

(71) The illustrated embodiment of the voltage regulator **700** employs an external DAC **750** and an associated reference voltage source **752**. The DAC **750** and the reference voltage source **752** take the place of an internal voltage reference for the error amplifier **704**, a feedback divider network, and a 1:1 buffer amplifier (which are described above in the context of the voltage regulators **500**, **600**). The DAC **750** and the reference voltage source **752** obtain operating power from a source other than the voltage that serves as the input to the voltage regulator **700**. For example, the DAC **750** and the reference voltage source **752** may be powered by the independent voltage supply and/or by another auxiliary voltage supply (not shown in FIG. **7**).

(72) The DAC **750** generates a fixed analog voltage that is equal to the desired regulator output voltage. The error amplifier **704** compares the voltage at the output voltage terminal **716** against the voltage applied to the reference voltage terminal **721**, and drives the series pass transistor **702** to maintain the minimum possible difference. The positive error input **736** of the error amplifier **704** has a very high impedance (as mentioned above with reference to the buffer amplifier **508**) to limit the amount of current drawn from the output voltage terminal **716** by the error amplifier **704**.

(73) The DAC **750** functions as a fixed reference voltage source, and its analog output is coupled to the reference voltage terminal **721**. The DAC **750** represents a digitally programmable component that can be configured to provide the desired reference voltage, up to a maximum voltage that is based on the voltage generated by the reference voltage source **752**. Thus, the reference voltage provided by the DAC **750** can be digitally programmed and adjusted to match the expected regulator output voltage present at the output voltage terminal **716**. In alternative implementations, a suitably configured analog voltage source or fixed voltage supply can be utilized instead of the DAC **750**. For example, if the regulator output voltage is expected to be 1.5 VDC, then an external 1.5 VDC voltage source can be coupled to the reference voltage terminal **721**.

(74) The basic operating principle of the linear voltage regulator **700** is similar to that described above for the voltage regulator **500**. The regulator output voltage (not a scaled version of it) is fed directly to the positive error input **736** of the error amplifier **704**, and the external DAC **750** provides the desired reference voltage to negative error input **732** of the error amplifier **704**, by way of the reference voltage terminal **721**. The error amplifier **704** generates its output based on the difference between the reference voltage and the regulator output voltage, and the output controls the impedance of the transistor to adjust the regulator output voltage as needed.

(75) While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or embodiments described herein are not intended to limit the scope, applicability, or configuration of the claimed subject matter in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the described embodiment or embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope defined by the claims, which includes known equivalents and foreseeable equivalents at the time of filing this patent application.

Claims

1. A linear voltage regulator comprising: an input voltage terminal configured to receive an input current; an output voltage terminal configured to output an output current substantially equal to the input current; a series pass field-effect transistor coupled between the input voltage terminal and the output voltage terminal; a voltage reference source to provide a reference voltage for the linear voltage regulator, the voltage reference source powered by an independent voltage supply that is isolated from the input voltage terminal; a buffer amplifier comprising a positive buffer input coupled to the output voltage terminal, and a negative buffer input coupled to a buffer output, the buffer amplifier powered by the independent voltage supply; and an error amplifier comprising an error output coupled to the series pass field-effect transistor, a positive error input operatively coupled to the output voltage terminal, and a negative error input coupled to the voltage reference source to receive the reference voltage, the error amplifier powered by the independent voltage supply, wherein the output of the error amplifier controls impedance of the series pass field-effect transistor to adjust a regulator output voltage at the output voltage terminal.
2. The linear voltage regulator of claim 1, further comprising a supply voltage terminal for the independent voltage supply, wherein: the buffer amplifier is coupled to the supply voltage terminal and to a ground terminal; the error amplifier is coupled to the supply voltage terminal and to the ground terminal; and the voltage reference source is coupled to the supply voltage terminal and to the ground terminal.
3. The linear voltage regulator of claim 1, wherein the series pass field-effect transistor is a metal-oxide-semiconductor field-effect transistor (MOSFET).
4. The linear voltage regulator of claim 3, wherein: the MOSFET has a source, a drain, and a gate; the source of the MOSFET is coupled to the input voltage terminal; the drain of the MOSFET is coupled to the output voltage terminal; and the gate of the MOSFET is coupled to the error output of the error amplifier.
5. The linear voltage regulator of claim 1, wherein the buffer amplifier is configured as a unity gain follower.
6. The linear voltage regulator of claim 1, wherein high input impedance of the positive buffer input isolates a current flow path from the input voltage terminal to the output voltage terminal, such that a ratio of current flowing into the positive buffer input to current flowing in the current flow path is less than 1:1000.
7. The linear voltage regulator of claim 1, further comprising a feedback divider network coupled between the buffer output and a ground terminal.
8. The linear voltage regulator of claim 7, wherein the positive error input is operatively coupled to the output voltage terminal via an output of the feedback divider network.
9. The linear voltage regulator of claim 7, wherein the feedback divider network comprises: a first resistor coupled between the buffer output of the buffer amplifier and the divider output; and a second resistor coupled between the divider output and the ground terminal.
10. The linear voltage regulator of claim 1, wherein the buffer amplifier comprises an input node

configured to receive an input current at least two orders of magnitude less than a magnitude of the output current at the output voltage terminal.
11. A test system for measuring electrical current consumption of a device under test (DUT), the test system comprising: a power capacitor configured to provide a capacitor voltage; a voltage regulator comprising a regulator input terminal and a regulator output terminal; a switching circuit to regulate electrical connections between a direct current (DC) voltage source, the power capacitor, and the voltage regulator; and a controller coupled to the power capacitor, the voltage regulator, and the switching circuit, the controller configurable to: control the switching circuit to place the test system into a charging state to charge the power capacitor with the DC voltage source; at a first time point, control the switching circuit to place the test system into a measurement state such that the power capacitor provides the capacitor voltage to the voltage regulator; detect, at a second time point after the first time point, by the controller sampling a voltage of the power capacitor and comparing the sampled voltage to a threshold, that loading by the DUT has caused the voltage of the power capacitor to drop below a threshold; and responsive to detecting that loading by the DUT has caused the voltage of the power capacitor to drop below the threshold, calculate electrical current provided by the power capacitor in a time period recorded during operation of the test system in the measurement state.
12. The test system of claim 11, further comprising a display device coupled to the controller, the display device controlled to display the calculated electrical current.
13. The test system of claim 11, the switching circuit comprising: a first switching element coupled between the DC voltage source and the regulator input terminal; and a second switching element coupled between the DC voltage source and a power terminal of the power capacitor, wherein: the controller keeps the first and second switching elements closed for the charging state to charge the power capacitor with source voltage provided by the DC voltage source, and to generate the DUT operating voltage from the source voltage; and the controller keeps the first and second switching elements open for the measurement state to provide the capacitor voltage to the regulator input terminal.
14. The test system of claim 13, wherein the controller is configurable to: record a measurement start time associated with opening of the first switching element when the second switching element is open, wherein the opening of the first switching element transitions the test system from the charging state to the measurement state if the second switching element is open; and record a measurement end time associated with closing of the first switching element when the second switching element is open, wherein the closing of the first switching element transitions the test system from the measurement state to a post-measurement state if the second switching element is open; wherein the time period is defined by the measurement start time and the measurement end time as recorded.
15. The test system of claim 11, wherein the electrical current is calculated based on a difference between a sampled value of a charged voltage of the power capacitor and a sampled value of a discharged voltage of the power capacitor.
16. The test system of claim 13, further comprising a diode having an anode coupled to the power terminal of the power capacitor, and having a cathode coupled to the regulator input terminal, the diode permitting discharge of the power capacitor when the first and second switching elements are open, and the diode inhibiting discharge of the power capacitor when the first switching element is closed and the second switching element is open.
17. The test system of claim 11, further comprising a calibration circuit couplable to the power capacitor to calibrate the capacitance of the power capacitor.
18. The test system of claim 11, further comprising: a first current-isolating buffer, coupled between the regulator input terminal and a first voltage input terminal of the controller, to isolate the controller from a test current flow path between the power capacitor and the DUT; and a second current-isolating buffer, coupled between a power terminal of the power capacitor and a second

voltage input terminal of the controller, to isolate the controller from the test current flow path.
19. The test system of claim 11, wherein the second time point corresponds to a time at which the power capacitor has reached a discharged voltage.
20. The test system of claim 11, wherein the first time point corresponds to a time at which the power capacitor has reached a charged voltage.